Activity 4.7 – Extended Run Counter

1.Design Approach:

Designed a 32-bit extended run counter using Verilog HDL to monitor valid transactions. Implemented synchronous reset and enable-controlled counting logic.

2.Verification Steps:

- Created a separate testbench.
- Applied reset.
- Activated the valid input signal.
- Tested multiple transaction cycles.
- Checked counter increment operation.
- Verified counter retention when valid is low.
- Simulated using ModelSim.

3.Results:

The testbench checks transaction counting and reset functionality. PASS/FAIL messages are displayed.

4.Issues Faced:

The counter must increment only when the valid signal is high. Reset conditions must be handled correctly. Counter width must support the required range.

5.Conclusion:

Successfully designed and verified an extended transaction counter using Verilog HDL.